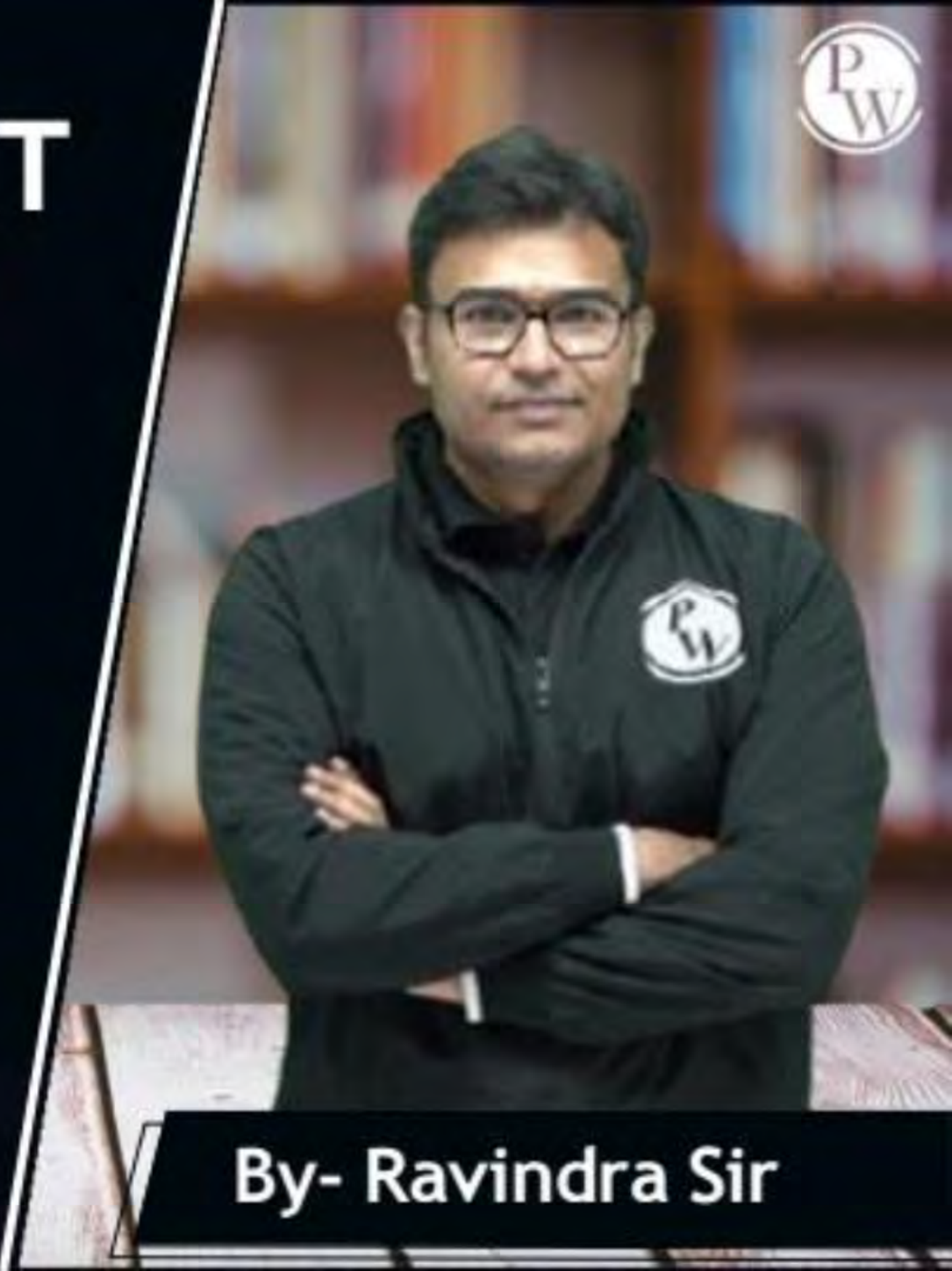
Computer Science & IT

COMPUTER NETWORK
(CN)

TCP and UDP

Lecture No. 02



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

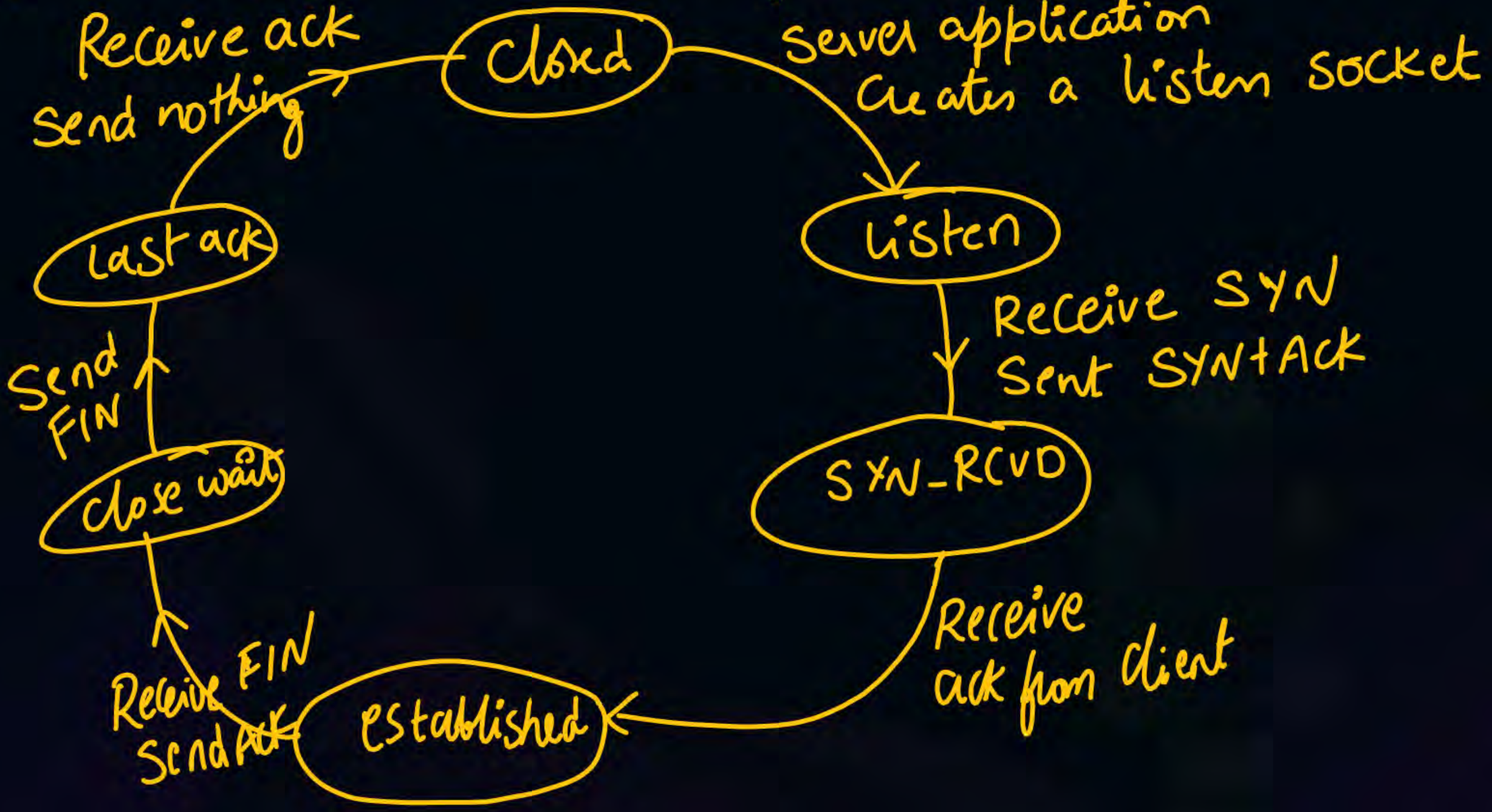
TCP

Topics to be Covered

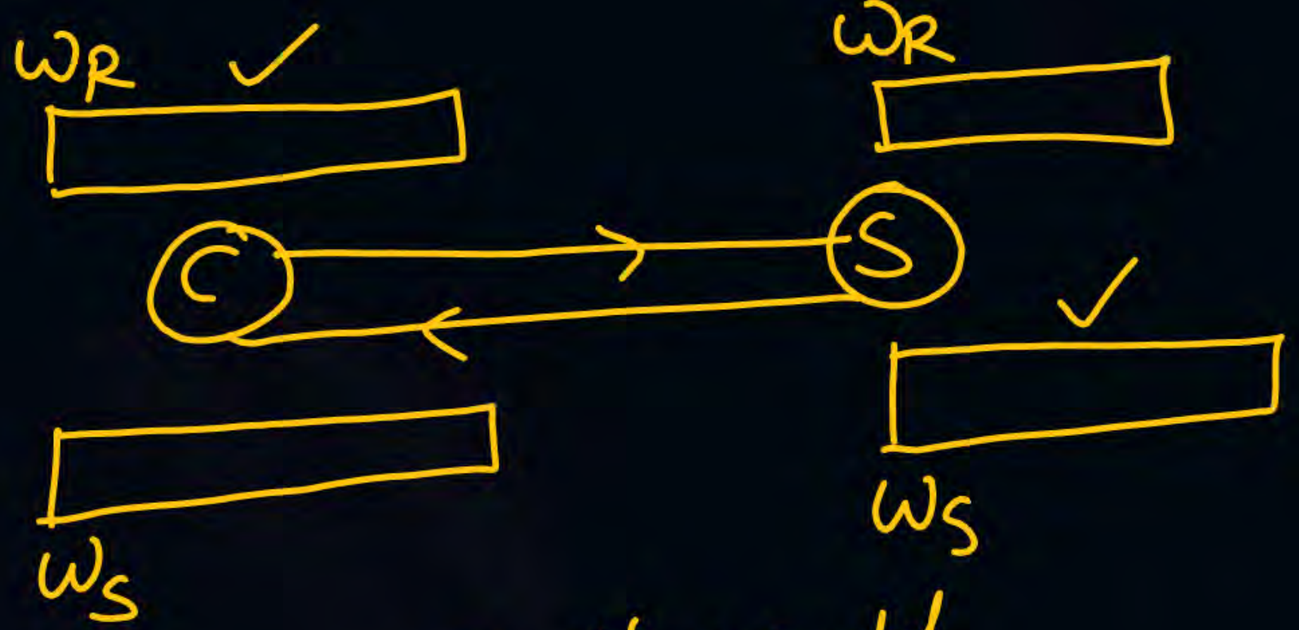


- Topic TCP
- Topic
- Topic

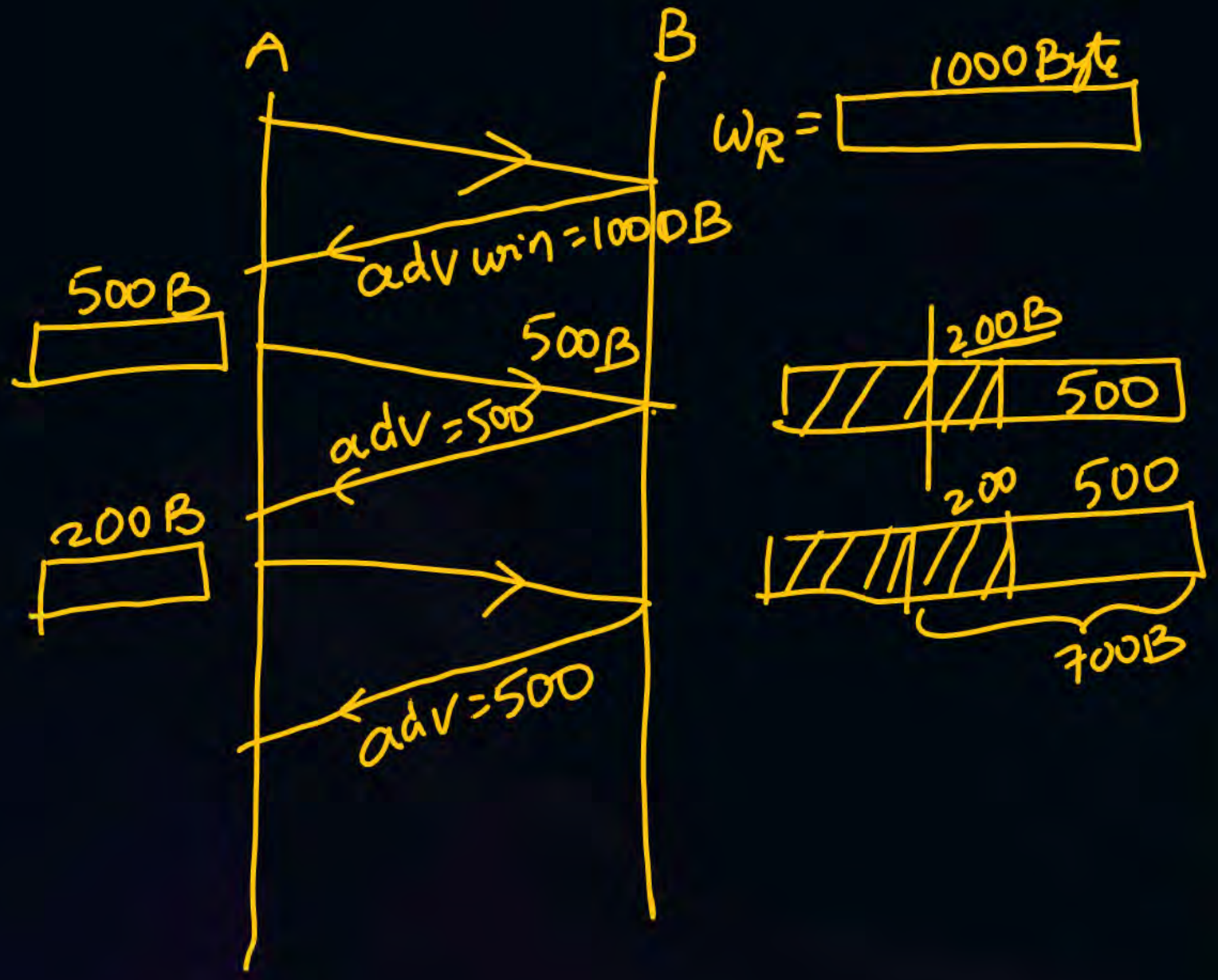
Sequence of states visited by Server in TCP

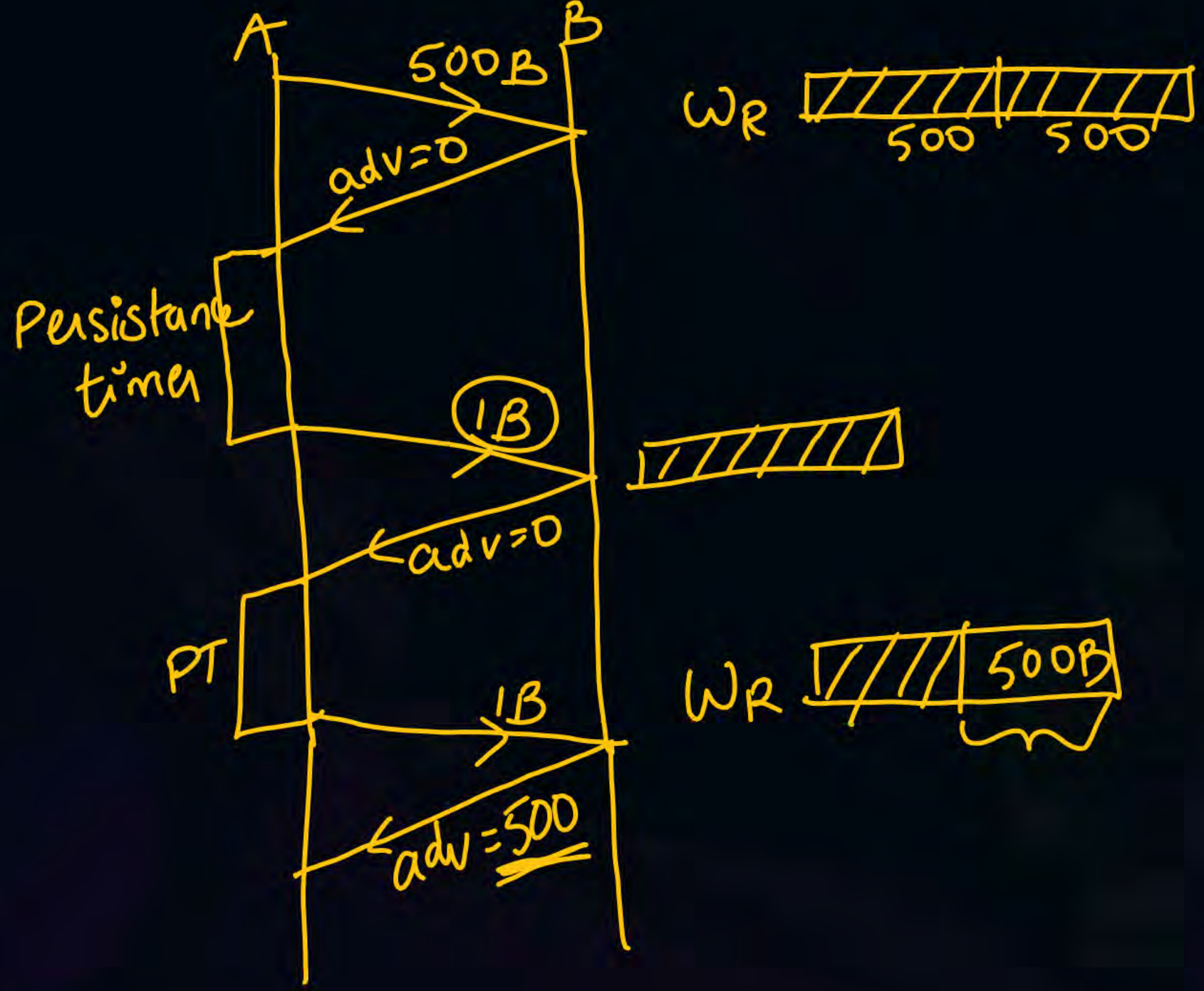


Window Size of advertisement window :-



For flow control


A
B
1000 Byte
W_R =
adv win = 1000B
500B
500B
adv = 500
200B
adv = 500
200B
500
200
500
700B


A
B
500B
adv=0
Persistance timer
1B
adv=0
PT
1B
adv=500
WR
500
500
WR
500B

One problem with adv window is it is only 16 bits

A receiver can only advertise 65535 max



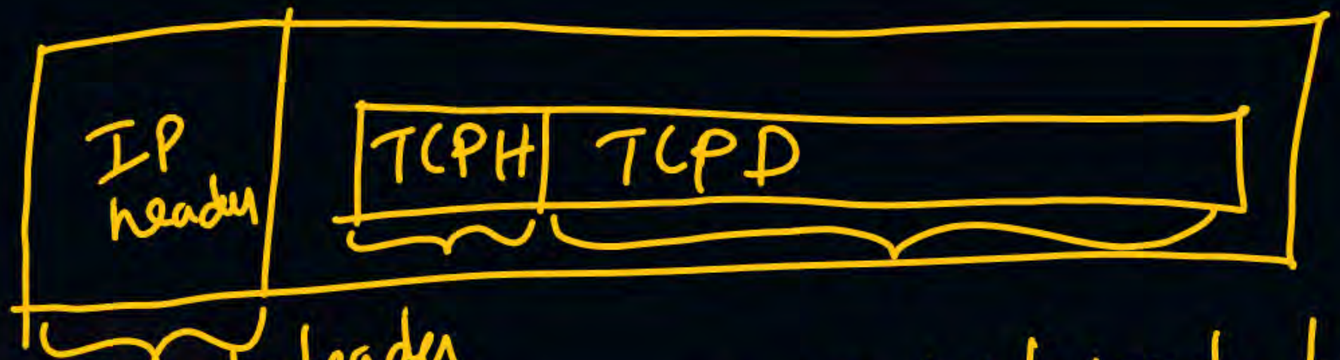
16 + 14

extra

↓

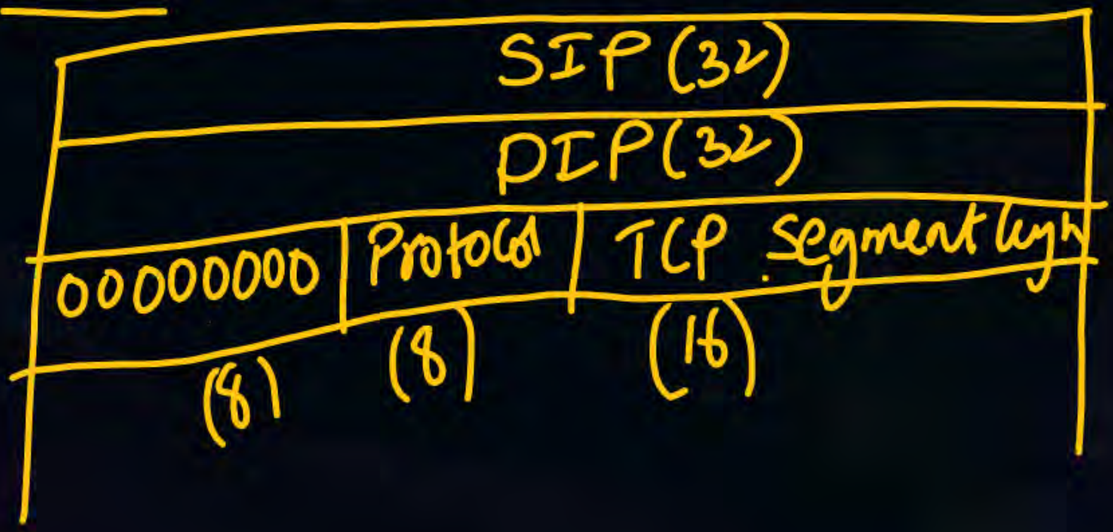
options.

Check Sum :-

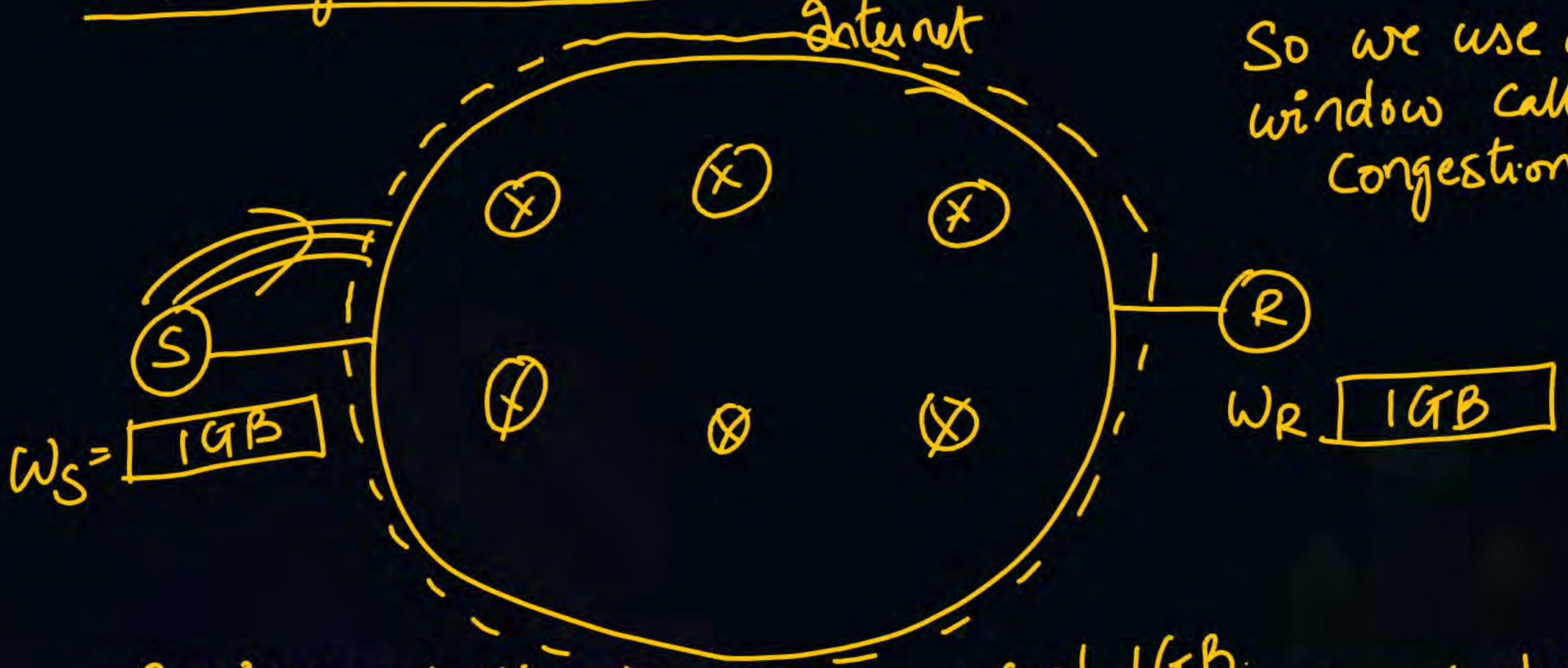


TCP CS → TCPH + TCPD + pseudo header from IP

PHIP



TCP Congestion Control :-



So we use one more window called Congestion window W_C

Receiver can receive 1GB, Sender can send 1GB
But the underlying n/w may not be in a position to handle 1GB

Sender will send $\min(W_S, W_C)$



adv win = 8kB

mss = 1kB

W_R = 8 seg

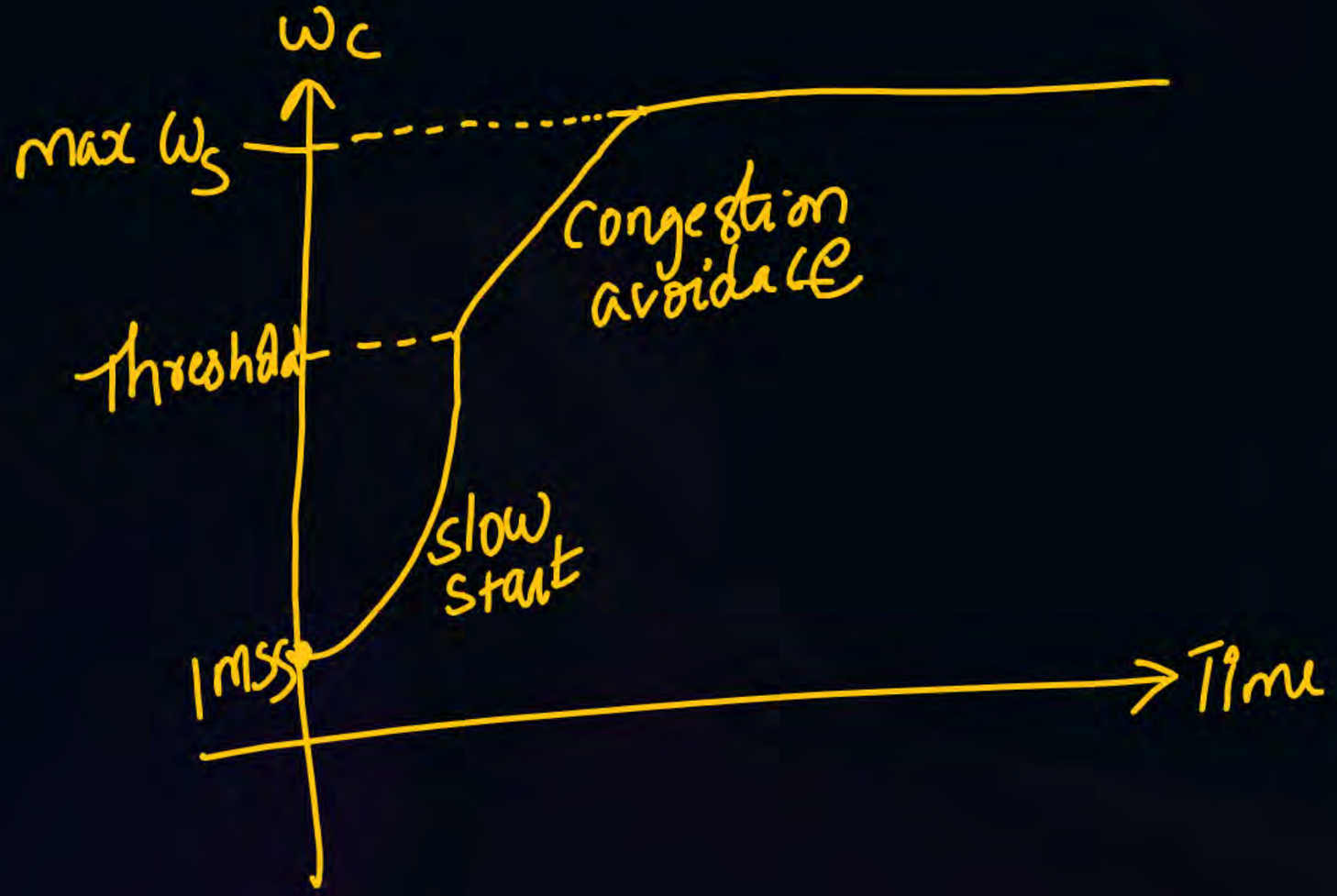
W_S = 8 seg ✓

W_C = 1 mss ✓

Threshold = $\frac{W_R}{2}$

1 | 2 | 4 | 5 | 6 | 7 | 8 ✓ = $\frac{8}{2}$ = 4 ✓

Congestion window growth



adv wind = 16 kB ✓

mss = 1 kB

w_R = 16 mss

th = 16/2 = 8 ✓

w_c = 1 mss

1	2	4	8	9	10	11	12	13	14	15	16 ✓

11 RTT

Retransmission in TCP:-

Retransmission after time out (TO)

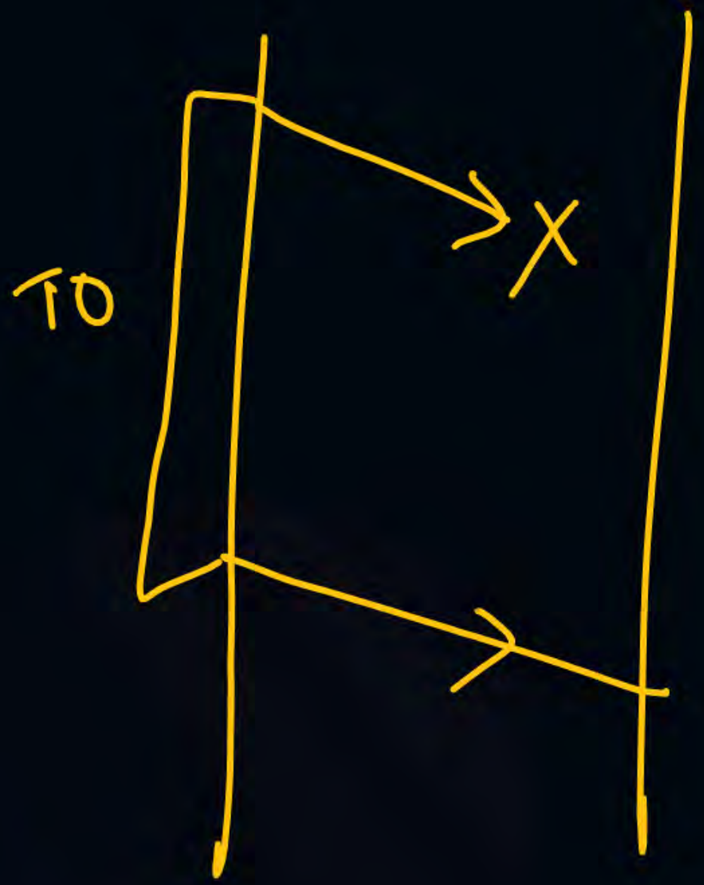


TCP uses → SR + GBN

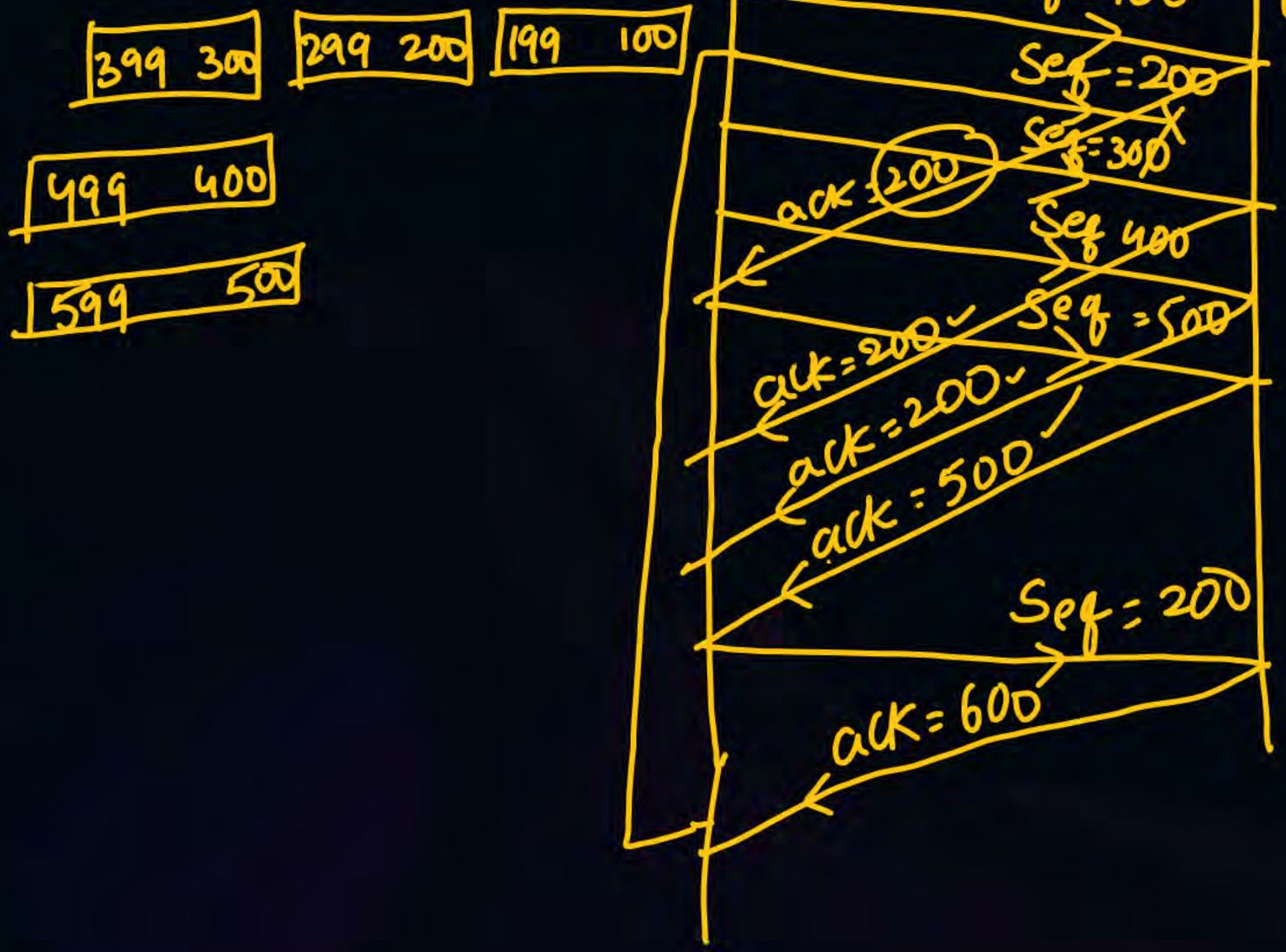
→ $w_S = w_R$ | Cummulative ACK

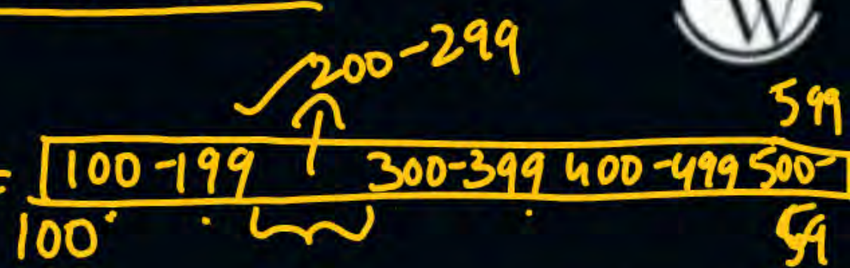
→ out of order packets are accepted

75% | 25%

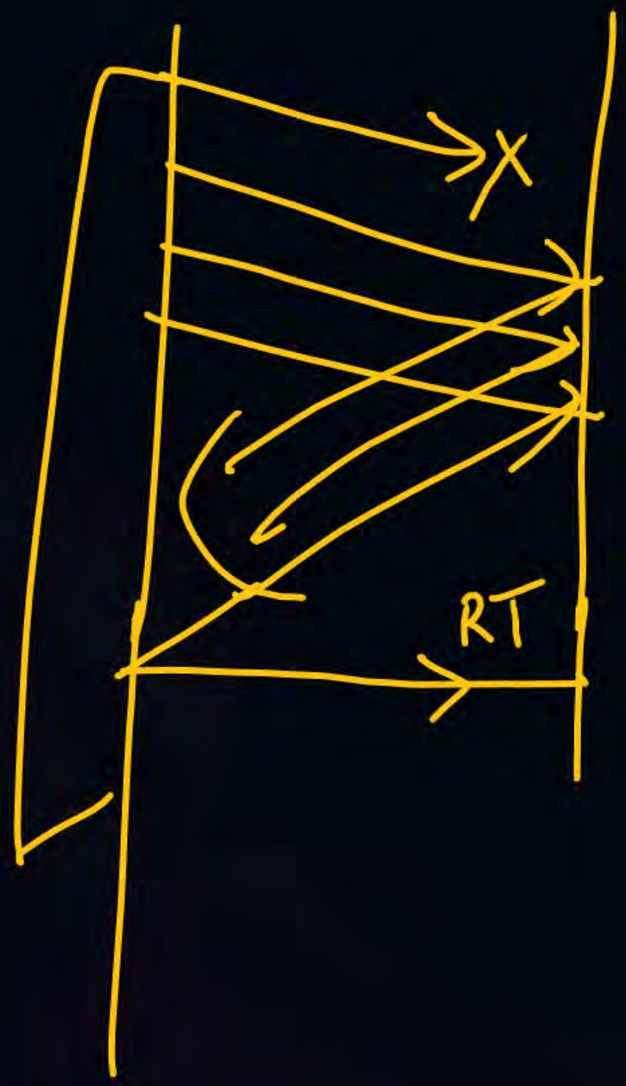

TO
X

RT after 3 duplicate acks & early retransmission :-



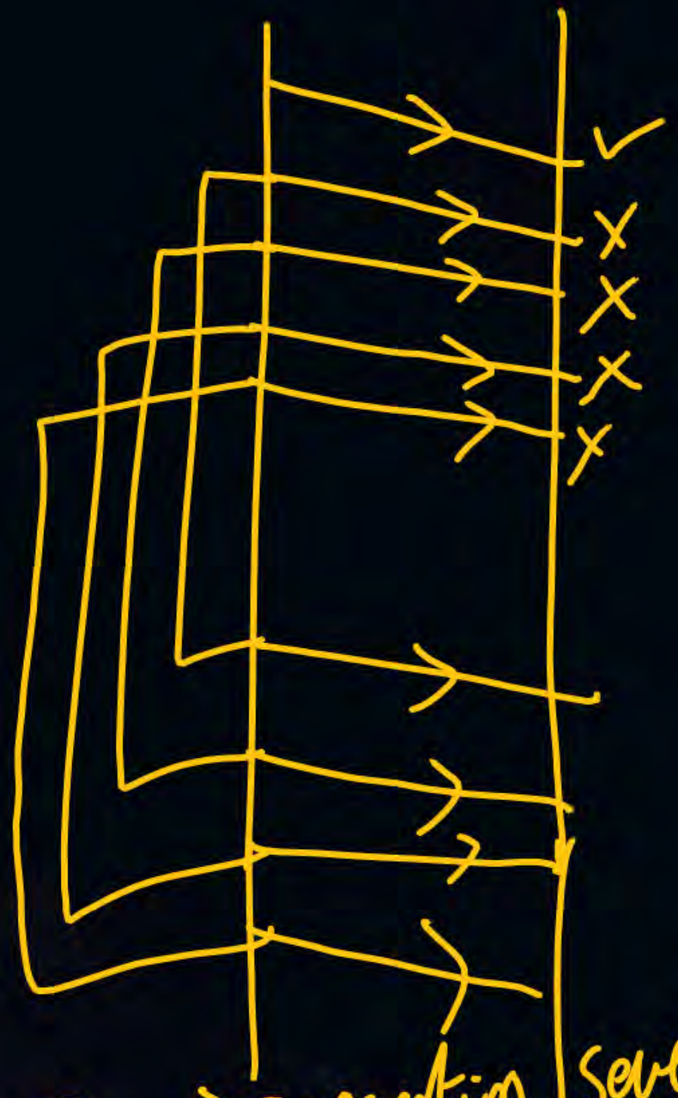


When Sender receives 3 dup acks, it is safe to assume one segment is lost


X
RT



3 dup ack → Congestion mild



TO → Congestion severe

Why is a packet lost?
Because of congestion

Congestion detection :-

1) TO timer → Severe Congestion
2) 3 dup ack → mild Congestion
X3) ICMP Source quench

Congestion Control algo

(i) Slow Start

(ii) Congestion avoidance

(iii) Congestion detection

a) TO

b) 3 Dup Ack

xc) ICMP Source quench

a) TO → Congestion severe

New Th = half of the current window Size

algorithm enters slow start

b) 3 dup packets → mild Congestion

New threshold 1/2 of current window Size

algorithm enters congestion avoidance phase

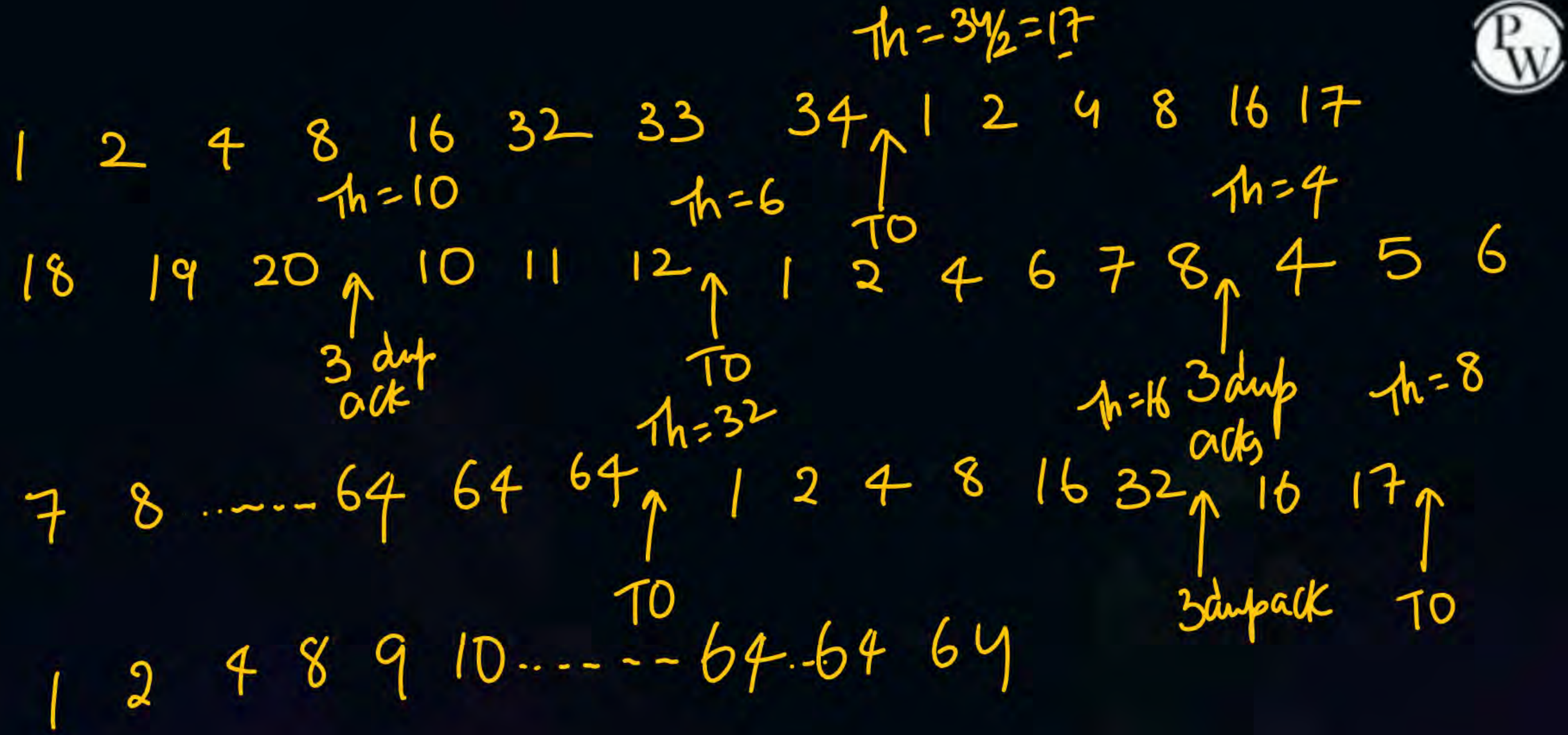
$W_R = 64\ KB$

$mSS = 1\ KB$

$W_S = 64\ KB = 64\ mSS$

$Th = W_S/2 = 32$

1 2 4 8 16 32 33 34 35 ..... 64 64 64 64 ...


Th=34½=17
1 2 4 8 16 32 33 34 1 2 4 8 16 17
Th=10
Th=6
TO
Th=4
18 19 20 10 11 12 1 2 4 6 7 8 4 5 6
3 dup ack
TO
Th=32
Th=16
3 dup acks
Th=8
7 8 ..... 64 64 64 1 2 4 8 16 32 16 17
TO
3dupack
TO
1 2 4 8 9 10 ..... 64 64 64

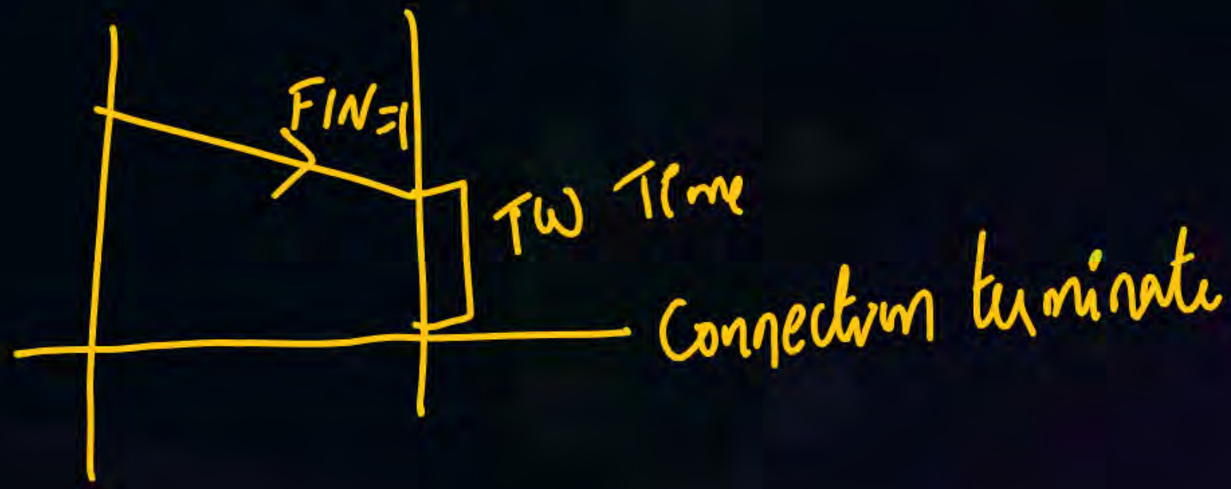
TCP timer management :-

1) Time wait timer
2) Keep alive timer
3) Persistent timer
4) ACK timer
5) Time out timer

Time wait timer :-



Whenever a host receives FIN packet, it will not close the connection immediatly, it will wait for few seconds to receive late packet



Keep alive timer



A server may have multiple connection and most of the connection are idle. This will waste server's resources

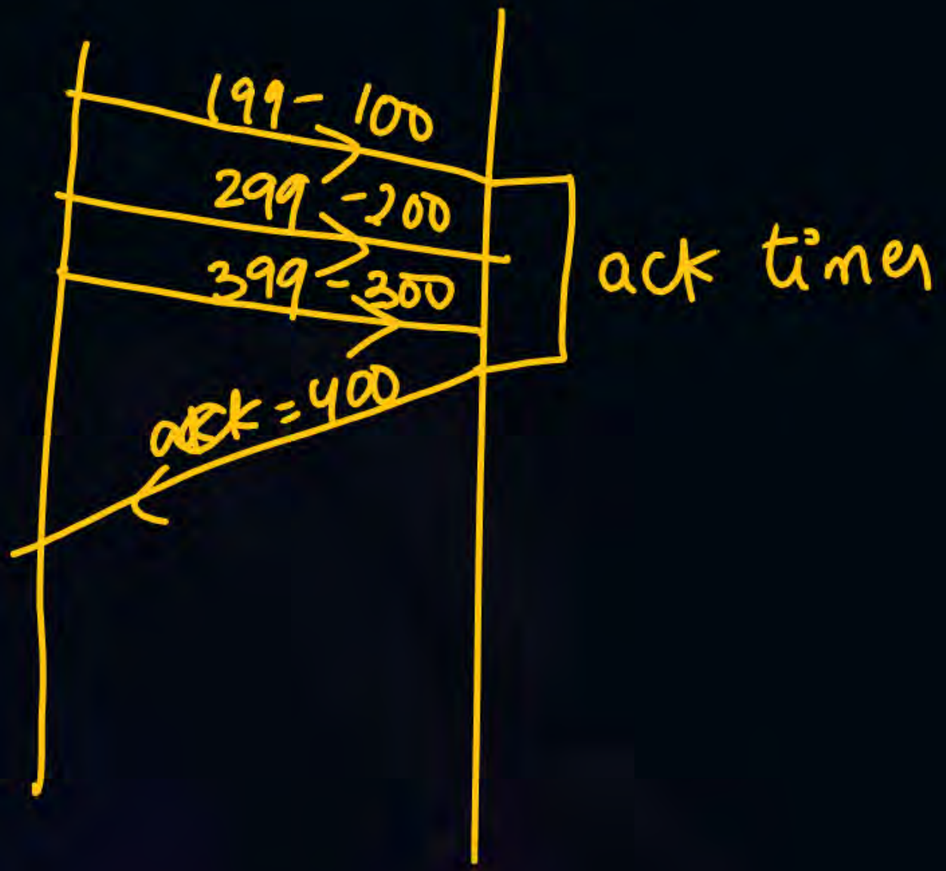


If there is no reply to probes, then server will terminate the connection

Persistent timer :-

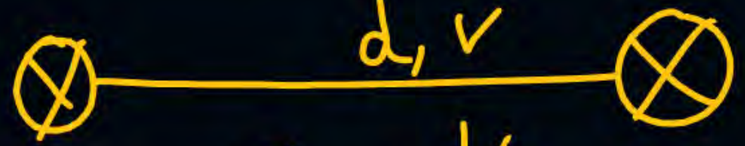


Acknowledgment timer :- (In TCP we use cummulative acks)



5) Time out time:-

In DLL only one link is considered at a time



$T_p = d/v$

$RTT = 2 * T_p$

Time out time $= 2 * RTT$ → Static

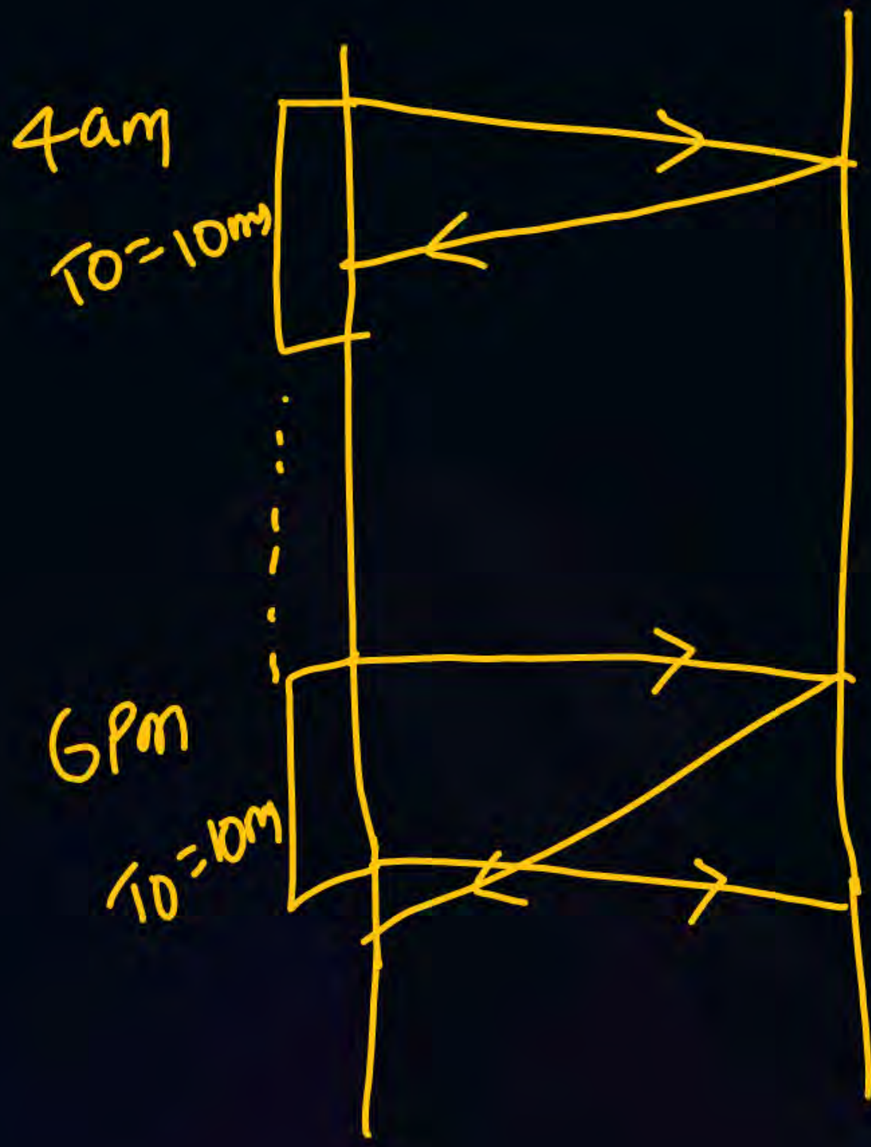
TCP → End to end



we cannot use static timer.

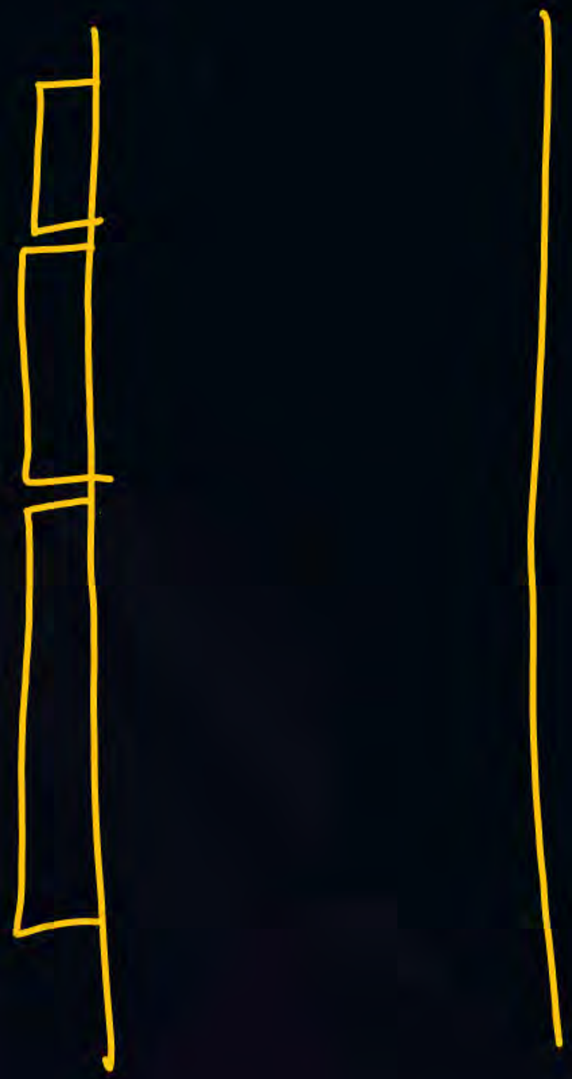
Because Route may change and traffic may change

So TO timer has to be dynamic.



⇒ leads to congestion

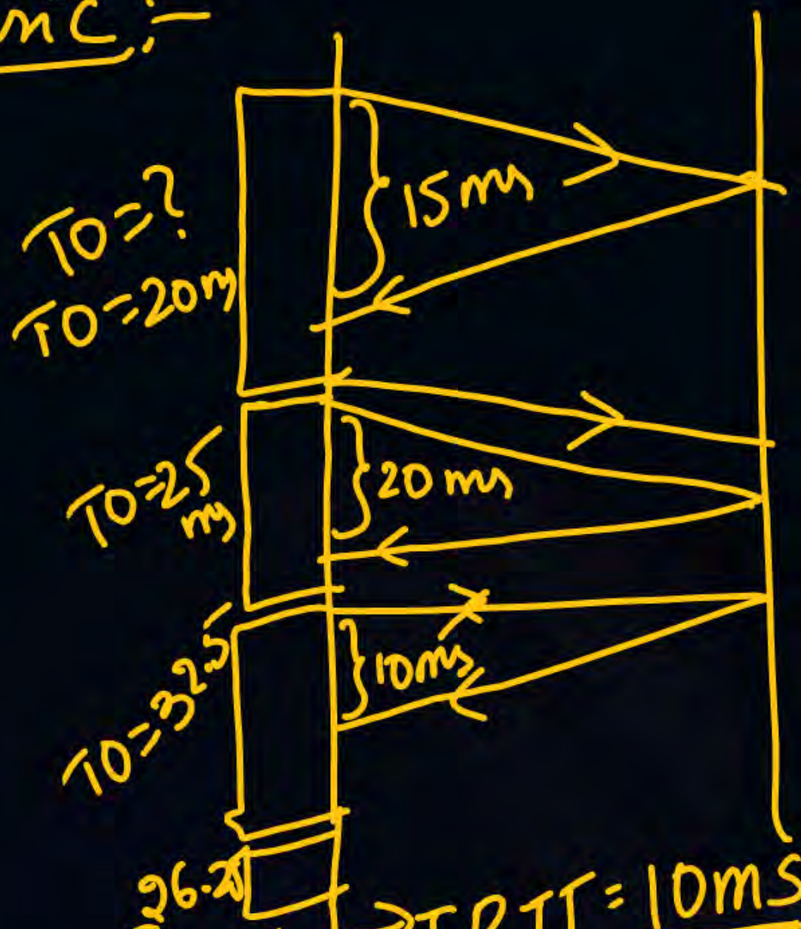
Dynamic
TO
time



There are two algorithms to find out TO time

1) Basic algo
2) Jacobson's algo

Basic :-



Guess → IRTT = 10ms ✓

TO time = 2 × RTT
= 20ms

ARTT = 15ms ✓

$NRTT = \alpha(IRTT) + (1-\alpha)(ARTT)$

$\alpha \rightarrow$ Smoothing factor

$0 \le \alpha \le 1$ (assume $\alpha = 0.5$)

$NRTT = 12.5\ ms$

$IRTT = 12.5\ ms$

$TO = 25\ ms$

$ARTT = 20\ ms$

$NRTT = \alpha(IRTT) + (1-\alpha)NRTT$

$= 16.25\ ms$

IRTT = 16.25 ms

TO = 2 × 16.25 = 32.5 ms

ARTT = 10 ms

NRTT = α(IRTT) + (1 − α) ARTT

= 13.125 ms

IRTT = 13.125 ms

TO = 2 * IRTT = 2 × 13.125 = 26.25

Dis adv of BA :-

Always TO = (2) × RTT

↓

No logic

Jacobson's algo will rectify this.

THANK - YOU